Sparse-View RGB-D 3D Reconstruction

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Team Setup

Student information

- Nguyễn Ngọc Minh – 20235533
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Presentation goal

Present the final RGB-D/source-depth reconstruction setting and the Run 30 source-ray correction result after the RGB-only learned extensions failed reconstruction-level gates.

Motivation

- 3D reconstruction is important because 3D representations help machines understand spatial structure, depth, and object relationships more accurately.
- Practical 3D capture often has only a small number of usable images.
- Classical high-quality pipelines depend on accurate camera poses or expensive scene-specific optimization.
- Sparse views increase ambiguity because cross-view correspondence is harder to establish reliably.
- A fast, generalizable method for unseen scenes would be useful for robotics, AR/VR, navigation, and scalable environment digitization.

Problem Statement

Core question

How can we reconstruct unseen indoor scenes accurately when only 2 to 5 sparse posed RGB-D views are available?

Key constraints

- Unseen scene at test time
- Sparse posed RGB-D views
- Known camera intrinsics/extrinsics
- No per-scene optimization

Expected outcome

- Robust geometry recovery
- Stable performance under view reduction
- Better occlusion and wrong-depth behavior
- Practical runtime per scene

Objectives

1. Build a strong sparse-view MV-DUSt3R+ baseline for 2–5 views.
2. Diagnose why RGB-only learned reliability, match filtering, and monodepth correction fail reconstruction-level gates.
3. Switch the final setting to posed RGB-D views with known camera intrinsics/extrinsics.
4. Validate Run 30 source-depth correction on overall, occlusion, and repeated/wrong-depth ambiguity subsets.

Final Hypothesis

Working hypothesis

Candidate deletion is too risky for sparse-view reconstruction. When input source depth is available, anchoring source rays to metric RGB-D geometry should correct wrong-depth candidates while preserving recall.

- Keep the useful MV-DUSt3R+ candidate set instead of over-filtering it
- Use source depth, poses, and intrinsics to correct 3D locations
- Test the result on occlusion and repeated/wrong-depth hard subsets

Proposed Contributions

Our contribution

The final contribution is a sparse-view RGB-D source-depth correction pipeline. RGB-only learned modules are kept as diagnostic evidence, while Run 30 is the accepted method.

- **Strong RGB-only baseline:** Run 11 improves B0 across 2–5 views using view selection and fixed confidence thresholds.
- **Negative RGB-only analysis:** Runs 22–29 show that proxy F1, candidate filtering, and generic monodepth do not solve reconstruction.
- **Final RGB-D method:** Run 30 uses input source depth maps with poses/intrinsics for source-ray correction and passes all gates.

Contribution Details

1. Baseline that solves sparse-view instability

- Run 11 fixed thresholds and view selection beat B0 on 2, 3, 4, and 5 views.
- This solves the first limit but leaves occlusion and wrong-depth ambiguity as hard cases.

2. RGB-only diagnostic path

- OARH, RSDH, RAJAH, and monodepth corrections are tested through reconstruction-level gates.
- They expose why filtering hurts recall/completeness and why generic monodepth is not enough.

3. Run 30 source-depth correction

- Uses source RGB-D depth maps with known camera poses/intrinsics.
- Corrects source rays in metric geometry instead of deleting candidates.

Why RGB-only Failed

Transition narrative

Runs 22–29 showed that RGB-only filtering/correction was insufficient. Run 30 changes the inference contract to RGB-D and passes the overall, occlusion, and ambiguity gates.

- Proxy F1 did not transfer to reconstruction-level F-score.
- Filtering removed valid sparse-view geometry and hurt recall.
- All-candidate retention became a strong non-learned baseline.
- RGB-only monodepth did not recover metric/source depth well enough.

Proposed Method

- Primary method: **MV-DUSt3R+**
- Single-stage feed-forward multi-view reconstruction
- Final setting uses sparse posed RGB-D views
- Run 30 adds source-depth correction after MV-DUSt3R+
- Our study focuses on robustness, hard cases, and runtime

Source: Tang et al., MV-DUSt3R+, project page and paper

Architecture Comparison

DUSt3R baseline

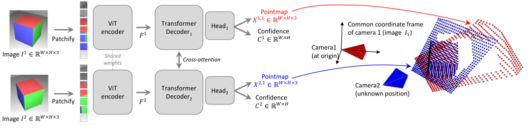
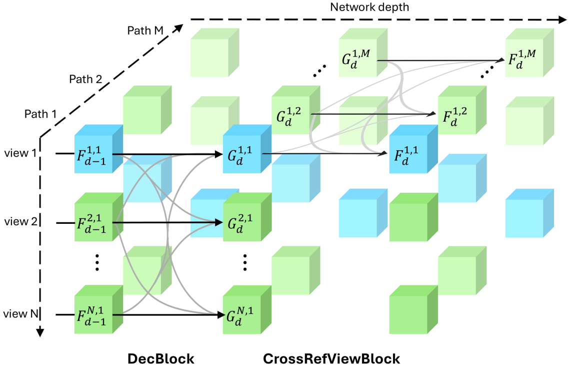


Figure 2. **Architecture of the network.** Two views of a scene (I^1, I^2) are first encoded in a Siamese manner with a shared ViT encoder. The resulting token representations F^1 and F^2 are then passed to two transformer decoders that constantly exchange information via cross-attention. Finally, two regression heads output the two corresponding pointmaps and associated confidence maps. Importantly, the two pointmaps are expressed in the same coordinate frame of the first image I^1. The network is trained using a simple regression loss (Eq. (4))

Pairwise reasoning with cross-attention between two views

MV-DUSt3R+ main method



Multi-path design with cross-reference-view information flow

Source: Left: Wang et al., DUSt3R, CVPR 2024. Right: Tang et al., MV-DUSt3R+, project page/paper

Why MV-DUSt3R+ vs DUSt3R?

Aspect	DUSt3R	MV-DUSt3R+
Processing style	Pairwise reconstruction	Joint multi-view reconstruction
Global consistency	Requires later alignment	Built into one forward pass
Sparse-view behavior	Can accumulate pairwise errors	More robust cue fusion
Runtime trend	Slower as views increase	Better scaling with more views
Role in project	Reference baseline	Main candidate method

- DUSt3R is still a strong baseline, so the comparison is direct and meaningful.
- MV-DUSt3R+ is the candidate generator for our final RGB-D correction step.

Quantitative Evidence from the Paper

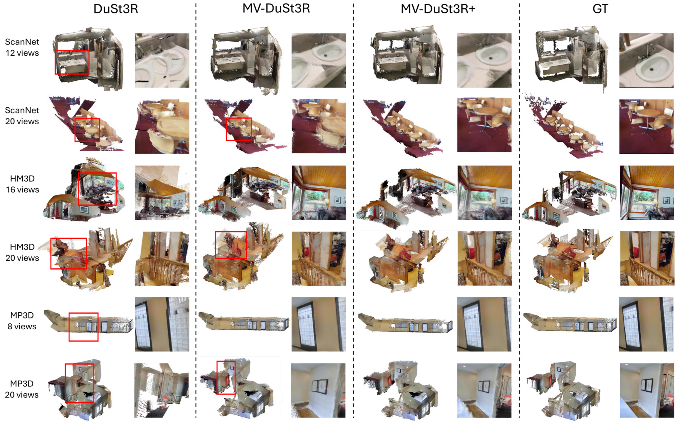
	Method	GO	HM3D ND ↓	HM3D DAc ↑	HM3D CD ↓	ScanNet ND ↓	ScanNet DAc ↑	ScanNet CD ↓	MP3D ND ↓	MP3D DAc ↑	MP3D CD ↓	Time (sec)
4 views	Spann3R	×	37.1	0.0	225(184)	8.9	19.5	54.7(50.1)	42.7	0.0	248(202)	0.36
	DUSt3R	✓	1.9	75.1	5.6(2.3)	1.3	89.8	4.0(0.4)	3.9	41.7	40.0(5.3)	2.42
	MV-DUSt3R	×	1.1	92.2	2.0(1.1)	1.0	93.3	2.0(0.4)	2.5	62.4	25.3(4.1)	0.05
	MV-DUSt3R+	×	1.0	95.2	1.5(0.9)	0.8	94.9	1.5(0.3)	2.2	68.0	19.9(3.4)	0.29
	$\text{MV-DUSt3R}_{\text{oracle}}$	×	1.0	94.6	1.5(0.7)	0.8	95.5	1.3(0.3)	2.3	66.6	20.7(4.0)	-
	$\text{MV-DUSt3R+}_{\text{oracle}}$	×	0.9	96.5	1.4(0.7)	0.7	95.8	1.2(0.2)	2.1	70.6	17.9(3.3)	-
12 views	Spann3R	×	32.6	0.0	125(113)	9.1	16.3	36.6(31.2)	35.0	0.0	138(112)	1.34
	DUSt3R	✓	3.9	30.7	18.1(3.4)	1.9	82.6	4.1(0.6)	6.6	12.0	49.6(8.3)	8.28
	MV-DUSt3R	×	1.6	79.5	3.0(1.2)	1.4	86.8	2.3(0.8)	3.4	41.3	22.6(5.5)	0.15
	MV-DUSt3R+	×	1.2	91.5	1.8(0.7)	1.2	88.4	1.8(0.7)	2.6	55.0	15.1(3.8)	0.89
	$\text{MV-DUSt3R}_{\text{oracle}}$	×	1.3	88.8	1.8(0.9)	1.0	90.6	1.3(0.7)	2.9	51.3	16.4(4.0)	-
	$\text{MV-DUSt3R+}_{\text{oracle}}$	×	1.1	94.8	1.4(0.7)	1.0	90.9	1.3(0.5)	2.5	59.8	13.6(3.5)	-
24 views	Spann3R	×	41.7	0.0	139(121)	11.4	1.6	37.4(35.5)	46.6	0.0	151(121)	2.73
	DUSt3R	✓	6.8	7.3	32.4(5.2)	2.4	72.6	5.1(1.0)	11.4	2.5	80.9(14.3)	27.21
	MV-DUSt3R	×	3.4	36.7	10.0(3.5)	2.2	75.2	2.7(0.9)	6.3	12.2	38.6(13.9)	0.35
	MV-DUSt3R+	×	2.1	64.5	3.9(2.0)	1.6	81.2	1.7(0.7)	4.3	26.7	22.0(5.9)	1.97
	$\text{MV-DUSt3R}_{\text{oracle}}$	×	2.1	58.9	3.5(2.1)	1.4	82.9	1.4(0.7)	4.4	22.0	19.9(5.1)	-
	$\text{MV-DUSt3R+}_{\text{oracle}}$	×	1.8	77.9	2.6(1.3)	1.3	85.1	1.3(0.6)	3.6	33.1	15.1(4.4)	-

Table 2 MVS reconstruction results. Best results are marked in red. For clarity, metric ND is scaled up by 10×, DAc is in percent %, and CD is scaled by 100× with its median given in parentheses. Results of our **oracle** methods are obtained by manually selecting the best single reference view from reference view candidates to report a possibly achievable

Across multiple datasets and view counts, the highlighted rows show that MV-DUSt3R+ improves reconstruction quality while keeping runtime practical.

Source: Tang et al., MV-DUSt3R+, Table 2

Qualitative Motivation



DUSt3R often fails on repeated structures, while MV-DUSt3R+ remains more stable in sparse-view scenes.

Source: Tang et al., MV-DUSt3R+, qualitative results

Final Pipeline

Select sparse posed RGB-D views and scene-level train / validation / test splits

↓

Run MV-DUSt3R+ to produce candidate point cloud and confidence

↓

Use source depth maps with poses/intrinsics to anchor source rays

↓

Apply Run 30 selective source-depth correction

↓

Evaluate overall, occlusion, and repeated/wrong-depth gates

Experiment Variables

Controlled factors

- Number of input views
- RGB-D source depth availability
- Scene-level train / validation / test split
- Fixed Run 30 policy on final test

Observed outcomes

- Reconstruction accuracy
- Completeness and consistency
- Runtime per scene
- Occlusion and repeated/wrong-depth failures

RGB-only Diagnostic Path

1. First build a supervised label cache with per-view visibility, occlusion, floating/wrong-depth, and match labels.
2. Mine occlusion-heavy, low-overlap, and hard-negative subsets before training the learned heads.
3. Freeze MV-DUSt3R+ and train small OARH / RSDH heads from generated labels, excluding direct target-label leakage features.
4. Runs 22–26 move the heads into reconstruction-level validation gates with fixed-confidence, top-k, and all-candidate baselines.
5. Run 27 is self-contained and learns a bounded confidence residual from actual candidates, self-geometry support, and raw image-patch cues.
6. Completed Run 27 still selects all candidates: learned filtering beats fixed confidence but loses completeness against full retention.
7. These runs motivate switching the final inference contract to RGB-D.

RGB-D Source-Depth Correction

1. Run 28 therefore preserves candidate count and learns source-ray 3D corrections from training depth and poses, using no depth at inference.
2. Completed Run 28 still fails the gate, but its source-depth oracle shows large occlusion and ambiguity headroom.
3. Run 29 approximates that oracle with RGB-only monocular depth aligned through the input poses and intrinsics.
4. Completed Run 29 regresses both occlusion and ambiguity, so RGB-only monodepth is not enough.
5. Run 30 makes source depth an explicit RGB-D input and gates full/selective source-ray correction policies.
6. Selected policy: `rgbd_residual_ge_0.30`, residual mode, $\alpha = 1.0$, threshold $0.30m$.
7. Final selected method: `rgbd_source_depth_selected`.
8. Keep final test clean: no threshold tuning, no checkpoint selection on test scenes.

Evaluation Data Used

Controlled ScanNet-style RGB-D subset

The reported Runs 19–31 use posed indoor RGB-D scenes with source images, depth maps, and known camera intrinsics/extrinsics.

Scene-level split

- 18 train scenes
- 6 validation scenes
- 6 held-out test scenes
- No test scene used for training or policy selection

Sparse-view protocol

- 3, 4, and 5 selected views
- Hybrid and diversity-aware view groups
- Geometry metrics at a 0.05 m threshold
- Hard occlusion and ambiguity subsets

Evaluation Scope

What the current evidence supports

Held-out comparison of Run 30 against all-candidate and fixed-confidence baselines on the controlled ScanNet-style subset.

What is not claimed

The current results are not a full ScanNet++ evaluation and do not establish official-benchmark generality. A larger mesh or laser-scan benchmark remains future work.

- Run 11 remains the strongest RGB-only baseline.
- Runs 12–29 are diagnostic and negative evidence.
- Run 30 is the final RGB-D/source-depth method.

Evaluation Plan

- Reconstruction F-score against available ground-truth geometry
- Accuracy, completeness, and Chamfer distance of point clouds
- Sparse 2–5 view behavior across scene splits
- Limit-specific gates for occlusion and wrong-depth ambiguity
- Runtime per scene for practical inference

	Method	GO	HM3D			ScanNet			MP3D			Time
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Source: Tang et al., MV-DUSt3R+, quantitative benchmark

Success criterion

Run 30 must beat the best non-learned baseline overall and improve both occlusion and ambiguity subsets by at least the gate margin.

Final RGB-D Result

Run 30 gate decision

`selected_method = rgbd_source_depth_selected`; best baseline = `all_candidates`; gate margin = 0.005; `pass_all_limits = 1`.

Split/subset	All cand.	Fixed conf.	RGB-D selected	Delta vs all
Val overall	0.1194	0.1123	0.1753	+0.0559
Val occlusion	0.1064	0.0917	0.2522	+0.1458
Val ambiguity	0.1623	0.1416	0.3000	+0.1377
Test overall	0.1758	0.1662	0.2764	+0.1007
Test occlusion	0.2111	0.1866	0.3146	+0.1035
Test ambiguity	0.1621	0.1462	0.2948	+0.1327

Why RGB-D Works

Final wording

RGB-only learned extensions did not pass reconstruction-level gates. After switching the inference contract to RGB-D, Run 30 uses input source depth maps with known camera poses/intrinsics for source-ray correction and passes the overall, occlusion, and ambiguity gates on held-out scenes.

- Source depth anchors each source ray to metric geometry.
- Correction preserves candidate count, unlike filtering that hurts recall and completeness.
- Occlusion and repeated/wrong-depth cases improve because the method fixes geometry rather than discarding sparse-view evidence.

Run 31 Coverage Stress Test

Purpose

Validate the frozen Run 30 method over more sparse-view groups. Run 31 does not train, tune, or select a new reconstruction method.

Coverage

- 30 scenes
- 12 groups per scene
- 360 groups total
- 3, 4, and 5 views
- No dense-frame inference

Fixed evaluation

- Hybrid and diversity-aware groups
- Two frame variants per configuration
- Frozen `residual_ge_0.30` policy
- Paired F-score delta vs all candidates
- Scene-cluster bootstrap 95% CI

Current status

Submitted/pending result. No additional stability claim is made yet.

Direct RGB-D Backprojection Baseline

Run 32 question

How well does direct depth backprojection reconstruct the same sparse-view groups without MV-DUSt3R+?

- Back-project valid depth with known intrinsics/poses and attach RGB.
- Apply fixed 2 cm voxel downsampling and cap at 3,500 points.
- Reuse Run 30 groups, metrics, and hard subsets; no policy tuning.

Required caution

Direct RGB-D backprojection is not source-depth correction. It is a depth-only/RGB-D baseline. Source-depth correction specifically refers to correcting MV-DUSt3R+ candidate points using source depth residuals.

Evaluator warning

GT uses the selected input depth maps, so this is not an independent mesh/laser-scan benchmark. Status: submitted/pending result.

Final Deliverables

- End-to-end implementation for Kaggle training/evaluation diagnostics
- Quantitative results on unseen scenes across multiple view counts
- Qualitative 3D visualizations and failure-case analysis
- Final report discussing RGB-D source-depth strengths and limitations
- Reproducible Kaggle experiment scripts organized under `scripts/kaggle`
- Final validation rerun script with a P100-compatible Torch fallback for unstable Kaggle GPU allocation
- Submitted supervised-extension kernels for Run 12–18 and Phase 3 label/subset kernels for Runs 19–20
- Submitted OARH v2 reconstruction path for Runs 21–23, exposing proxy-to-geometry transfer limits
- Submitted RSDH v2 reconstruction path for Runs 24–26, where proxy match validity fails the geometry gate
- Redesigned Run 27 as a reconstruction-aware ensemble; its gate selects all-candidate retention over learned filtering
- Added Run 28 source-ray correction to repair wrong depth without deleting valid occluded geometry
- Added Run 29 monodepth source-ray correction to approximate the Run 28 oracle without GT depth at inference
- Added Run 30 RGB-D source-depth correction as the final accepted solution for the two remaining limits
- Added Run 31 as a frozen-policy coverage stress test over 360 sparse-view groups; results are pending
- Added Run 32 as a direct RGB-D backprojection baseline without MV-DUSt3R+; comparison with Run 30 is pending
- Clean GitHub repository with curated docs, proposal sources, and PDF deck

Final Contributions

- A strong RGB-only baseline that solves sparse-view instability through view selection and fixed confidence thresholds.
- A negative-analysis record showing why OARH/RSDH/RAJAH and monodepth should not be claimed as RGB-only solutions.
- A final RGB-D/source-depth method that passes reconstruction-level gates for overall quality, occlusion, and repeated/wrong-depth ambiguity.
- A clear limitation boundary: RGB-only is baseline/diagnostic; Run 30 is the final source-depth contribution.

References

- Wang et al., “DUSt3R: Geometric 3D Vision Made Easy,” CVPR 2024.
- Tang et al., “MV-DUSt3R+: Single-Stage Scene Reconstruction from Sparse Views In 2 Seconds,” CVPR 2025.
- Yeshwanth et al., “ScanNet++: A High-Fidelity Dataset of 3D Indoor Scenes,” ICCV 2023.
- DTU Robot Image Data Sets, Multi-View Stereo Benchmark.